

Ch 5: Long Short-term memory (LSTMs)

VI: Why RNNs (Recurrent Neural Network)

1) If we use BOW, TFIDF, W2V we are disregarding the sequence of words.

But sequence is very important.

2) Time Series data

We break time into intervals and we take those window values.

OR

we take frequency domain with Fourier Transform.

3) Machine Translation Problem

English $\xrightarrow{\text{ML model}}$ Spanish
Sequential Sequential.

So these models are called Sequence 2 Sequence models.



Date: _____

Page No.: _____

4) Speech Recognition (Siri, Alexa, Google Assistant)

Seq. in audio $\rightarrow$ Sentence in English.

5) Image $\rightarrow$ Captions { English Sentence }

Core idea $\rightarrow$ If your output depends on seq. of inputs then a new type of NN is required.

eg $\rightarrow$ Amazon Food Reviews

Sentence $\rightarrow$ 2-class classifier
 x_i y_i

$x_i \rightarrow x_{i1}, x_{i2}, x_{i3}, x_{i4} \dots x_{i5}$

Take each $x_{i1}, x_{i2} \dots$ as one hot encoded vector.

Think if an MLP could be trained on it or not.

It would be very lengthy with that approach.

Because there might be billions of weights.

So RNN was designed.

V2% Recurrent Neural Network

↳ Repeats

eg → Amazon Food Reviews (Binary Classification Task)

$$D = \{x_p, y_p\}$$

Sequence of words $\rightarrow$ binary

$$y_0 \leftarrow x_1 \rightarrow x_{11}, x_{12}, x_{13}, x_{14}, x_{15}$$

$$x_2 \rightarrow x_{21}, x_{22}, x_{23}$$

No. of words are different in each sentence.

$$x_p \rightarrow x_{p1}, x_{p2}, x_{p3}, \dots, x_{p, m_p}$$

$$M_p = \text{len of } x_p$$

words are one-hot encoded.

$$\text{Size of vocab} = d$$

words are one-hot encoded

$x_{0,j} \in \mathbb{R}^d$
 $\downarrow$
 $\rightarrow \text{bin-vec}$

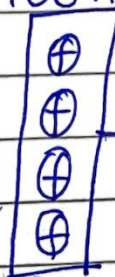
Task

To classify a sequence of words to 0 or 1.

$x_1, \langle x_{11}, x_{12}, x_{13}, x_{14} \rangle$ (y)

At Time = 1 (1) x_{11}

100 neurons



$o/p_i \in \mathbb{R}^{100}$
 $\rightarrow$

Also a vector

Hidden layer

Weight (w)

$x_{11} \in \mathbb{R}^d$

1/p layer

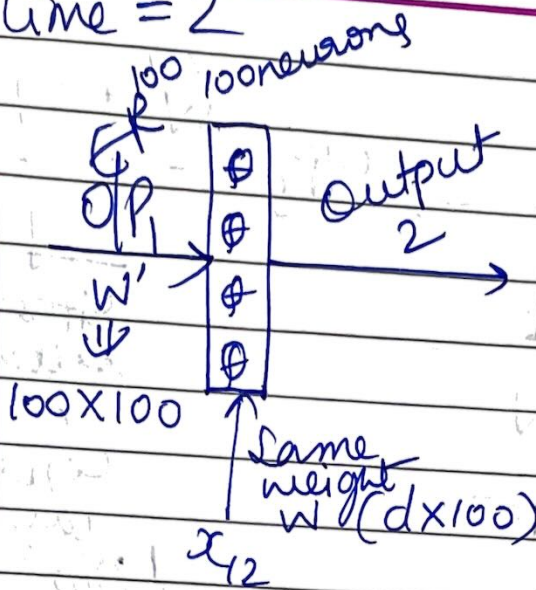
q

W : matrix ($d \times 100$)

$o/p_i = f(Wx_{11})$

f = either tanh / Sigmoid / ReLU.

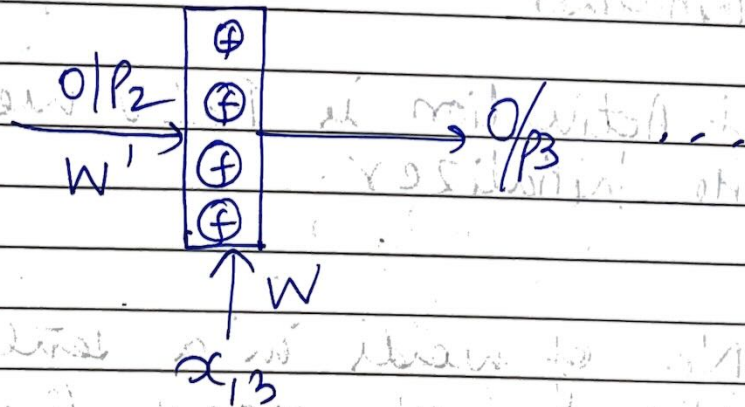
At Time = 2



we take O/P_1 as i/p to 2nd layer
we want to maintain the sequence information.

$$O/P_2 = f(Wx_{i2} + W'O/P_1)$$

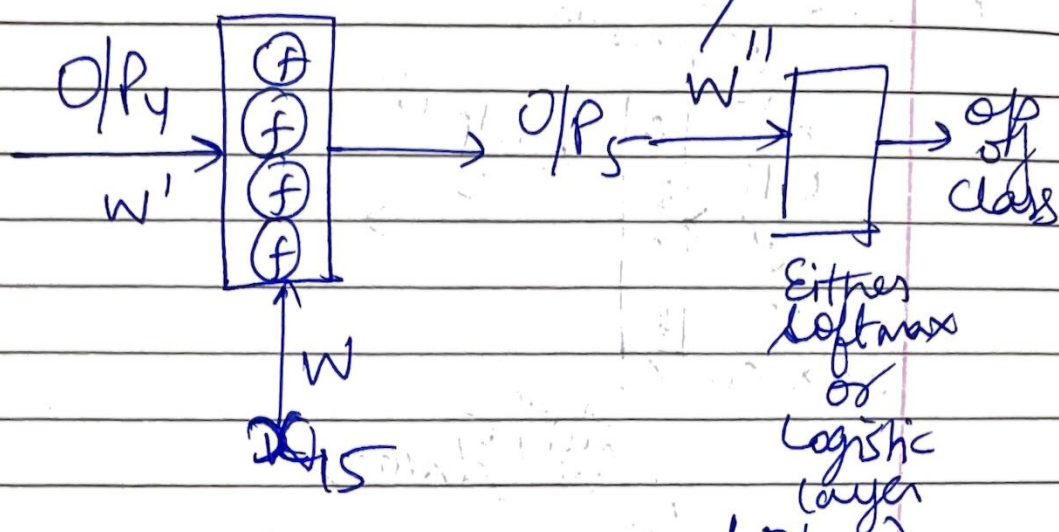
Time = 3



till
 $T=5$
 (since
 5 words
 in
 sent 1)

$$O/P_3 = f(Wx_{i3} + W'O/P_2)$$

At
T=5



$$O/P_5 = f(Wx_{15} + W'O/P_4)$$

Note :-

Initializers of weights depend on type of activation unit.

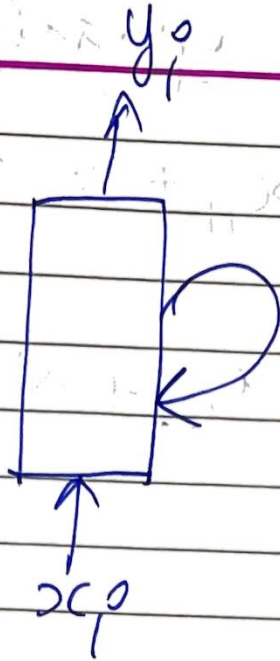
1) If Activation is Sigmoid or tanh we have Glorot or Xavier initializer

2) If Activation is ReLU we have He initializer

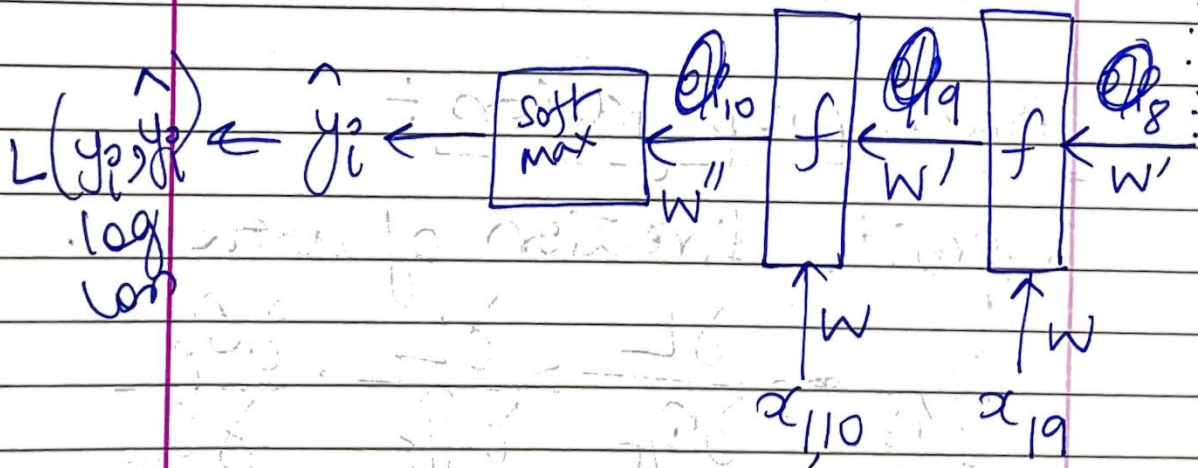
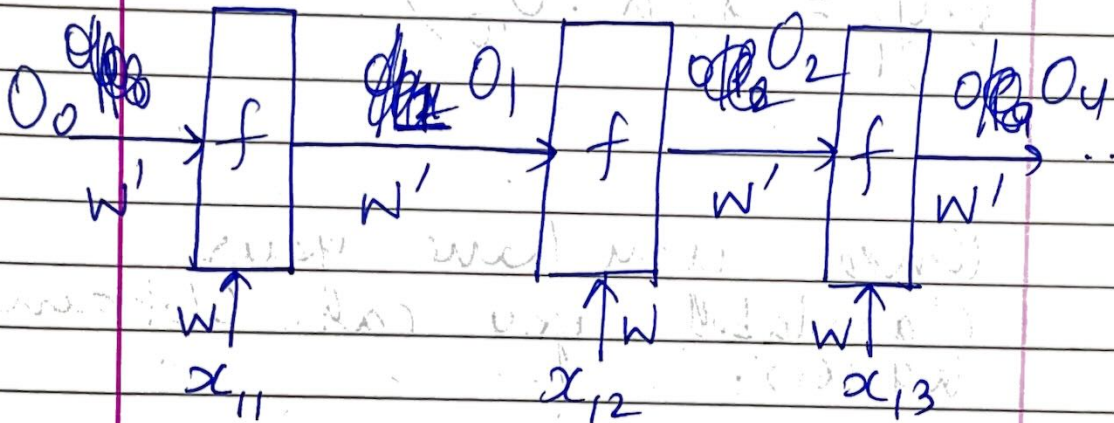
No. of words in a sentence till how many time layer on the time axis it needs

note If 3 words we do it till time=3
" 5 " " " " " =5

$\hat{y}_0$
ification.



V3: Training RNNs: Backprop



Time-axis

Forward Propagation



Date :

Page No. :

$$O_1 = f(wx_{11} + w'O_0)$$

$$O_2 = f(wx_{12} + w'O_1)$$

$$\vdots$$
$$O_{10} = f(wx_{1,10} + w'O_9)$$

$$\hat{y}_i = g(w'' \cdot O_{10})$$

Once you have your class label you can obtain log loss.

Back propagation :

opposite direction of arrows.

$$\frac{\partial L}{\partial \hat{y}_i} = \frac{\partial L}{\partial O_{10}} \cdot \frac{\partial \hat{y}_i}{\partial O_{10}}$$



Date :

Page No. :

$$\frac{\partial L}{\partial w''} = \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial w''}$$

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial O_{i0}} \cdot \frac{\partial O_{i0}}{\partial w}$$

In backprop, when reached 1st word

$\frac{\partial L}{\partial w}$ - lots of multiplication of partial derivative which is less than 1.

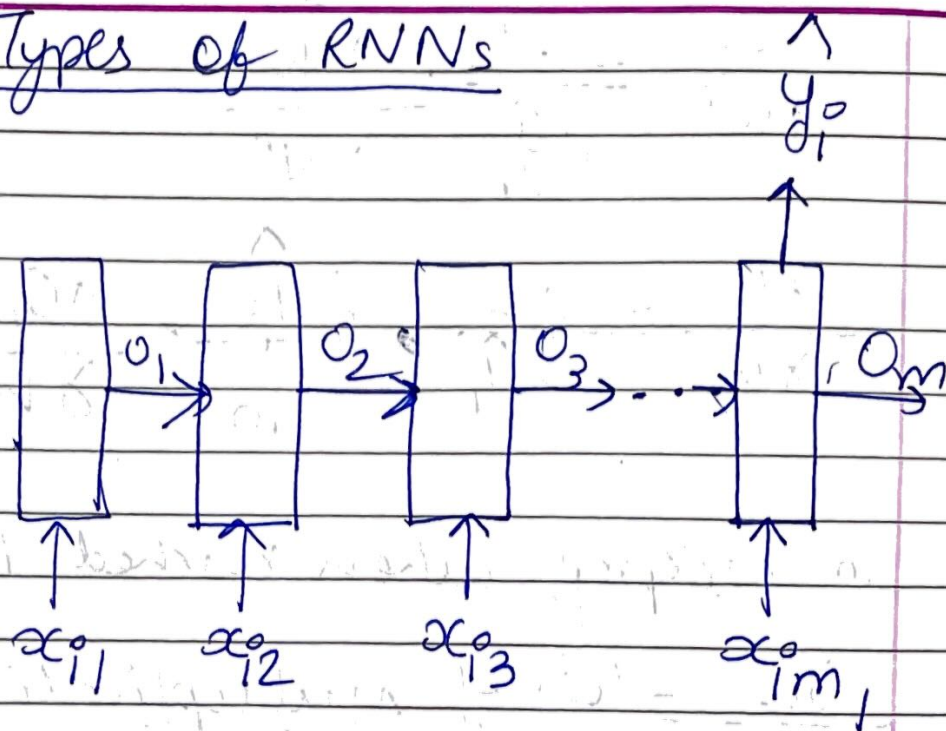
So we get into vanishing gradient problem.

We can also have exploding gradient problem when all/some partial derivatives are greater than 1.

It happens $\because$ the sequence of words are long.

Modifications of RNN are LSTM and GRU.

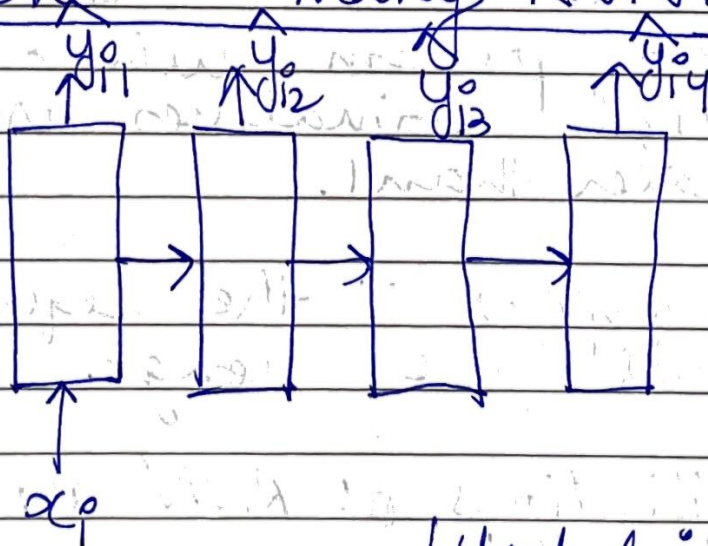
V4: Types of RNNs



Many-to-One RNN

Good ^{for} Useful for Sentiment Classification

One-to-many RNN:-



(no sequence in the input)

Useful in Image captioning



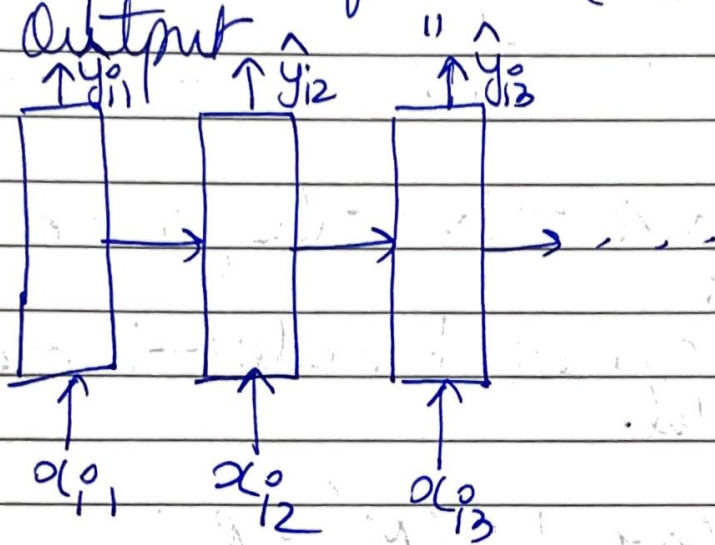
Date :

Page No. :

Many-to-many $\begin{cases} \text{Type A} \\ \text{Type B} \end{cases}$

Type A :

Input sequence (Same length)

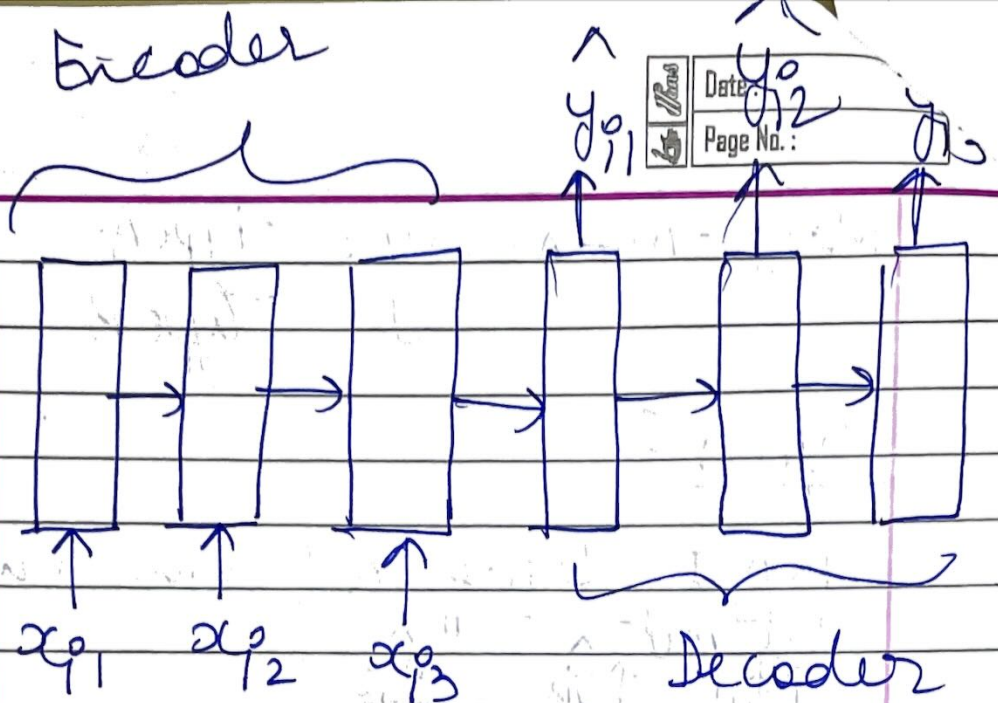


eg $\rightarrow$ Useful in parts of speech detection

Type B :

Many-to-many RNN
different length

eg $\rightarrow$ Machine Translation
English to Spanish
 $(20) \rightarrow (10)$
 $(10) \rightarrow (20)$

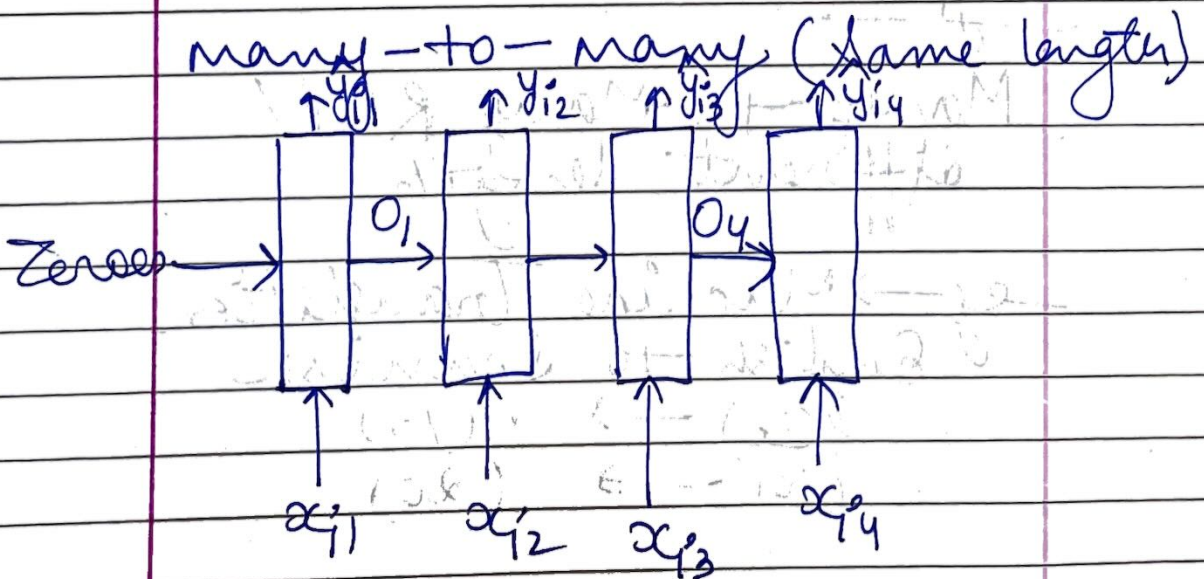


Sequence - to - Sequence
model

Also called Encoder-Decoder
model.

V5: Need for LSTM/GRU

What is the problem with simple
RNN?



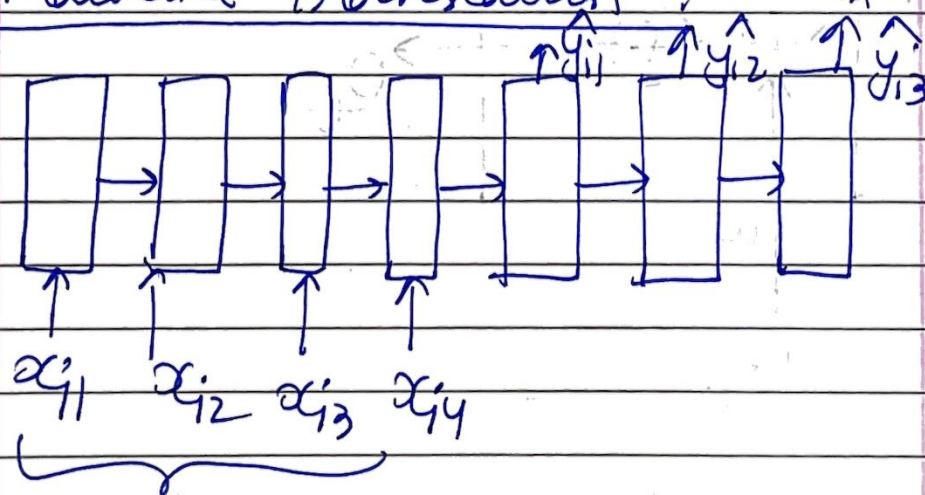
$\hat{y}_{i4} \rightarrow$ depends a lot on x_{i4} & o_3
 $\hat{y}_{i4} \rightarrow$ depends less on x_{i1} & o_1

In real world

$y_{i4} \xrightarrow{\text{more}} x_{i1}$
 less on x_{i4}

Simple RNN's can't take up long term dependencies (if a later o/p depends on earlier input, it is called long term dependencies).

Machine Translation



Eng

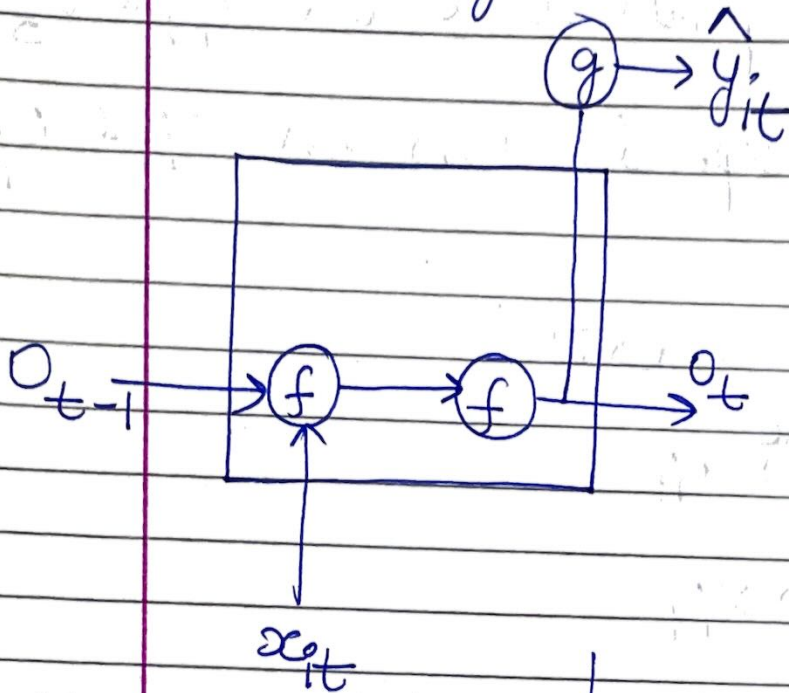
$(x_{i1}, x_{i2}, x_{i3}, x_{i4}) = \hat{y}_{i1}$
 $(x_{i2}, x_{i3}, x_{i4}) = \hat{y}_{i2}$
 $(x_{i3}, x_{i4}) = \hat{y}_{i3}$



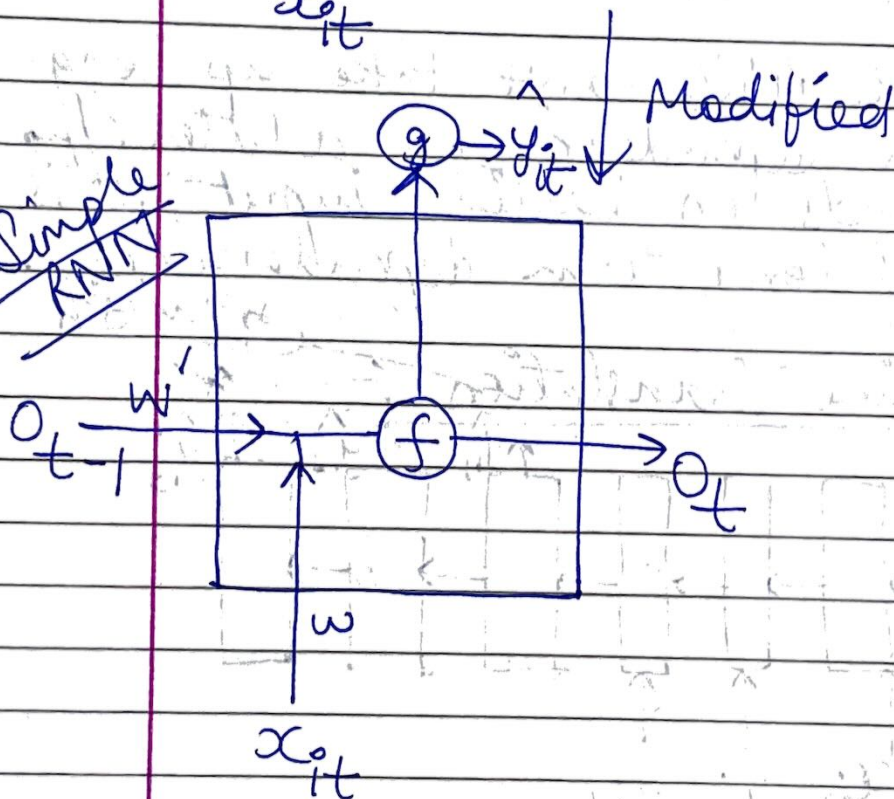
Date :

Page No. :

V6: LSTM (Long Short Term Memory RNN)



~~Simple RNN~~



$$o_t = f(wx_{it} + w'o_{t-1})$$

$$\hat{y}_{it} = f(w''o_t)$$

Doubtful.



Date :

Page No. :

$$O_t = f([w', w] \cdot [x_{it} \ O_{t-1}])$$

$$\hat{y}_{it} = g(w'' O_t)$$

Colah blog (Christopher Olah)

Notation

$$c_t = O_t$$

$$h_t = y_{it}$$

↓
Cell state

Pointwise operation is not a layer

X multiply $\rightarrow$ If vector all 1's then info just copied.

$\rightarrow$ If vector all $1/2$'s then info is forgotten by LSTM.

+ Sum $\rightarrow$ How much new info should be added.

W_f = Forget gate
Concatenate operation
bias.

* $\rightarrow$ Pointwise product.



Date : _____

Page No. : _____

V7: GRUs (Gated Recurrent Unit)

LSTM (1997) — input gate
~~for~~ output gate
forget gate.

GRU (2014) — Simplified version
inspired from LSTM.

- faster to train.
- As powerful as LSTM
- Reset and Update gates.
- Lesser equations, lesser partial ~~der~~ derivatives. Hence faster computations. Because of faster backpropagation.

V8 Deep RNN

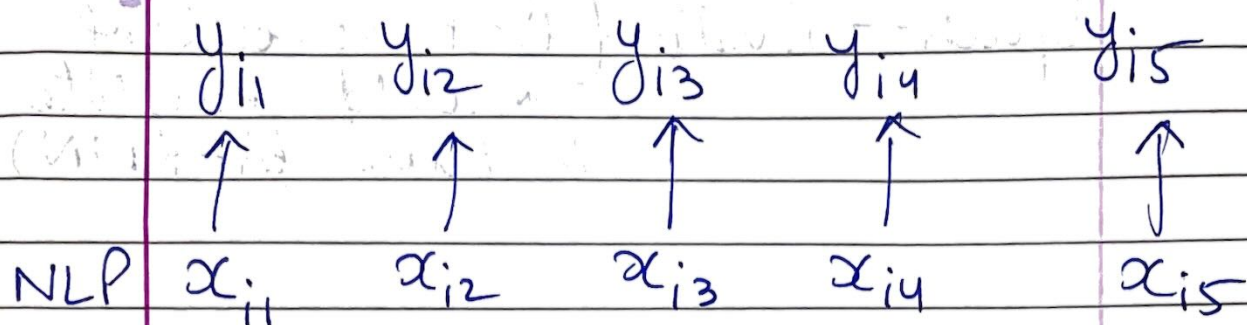
Actual structure of deep RNN is actually dependent upon the problem.



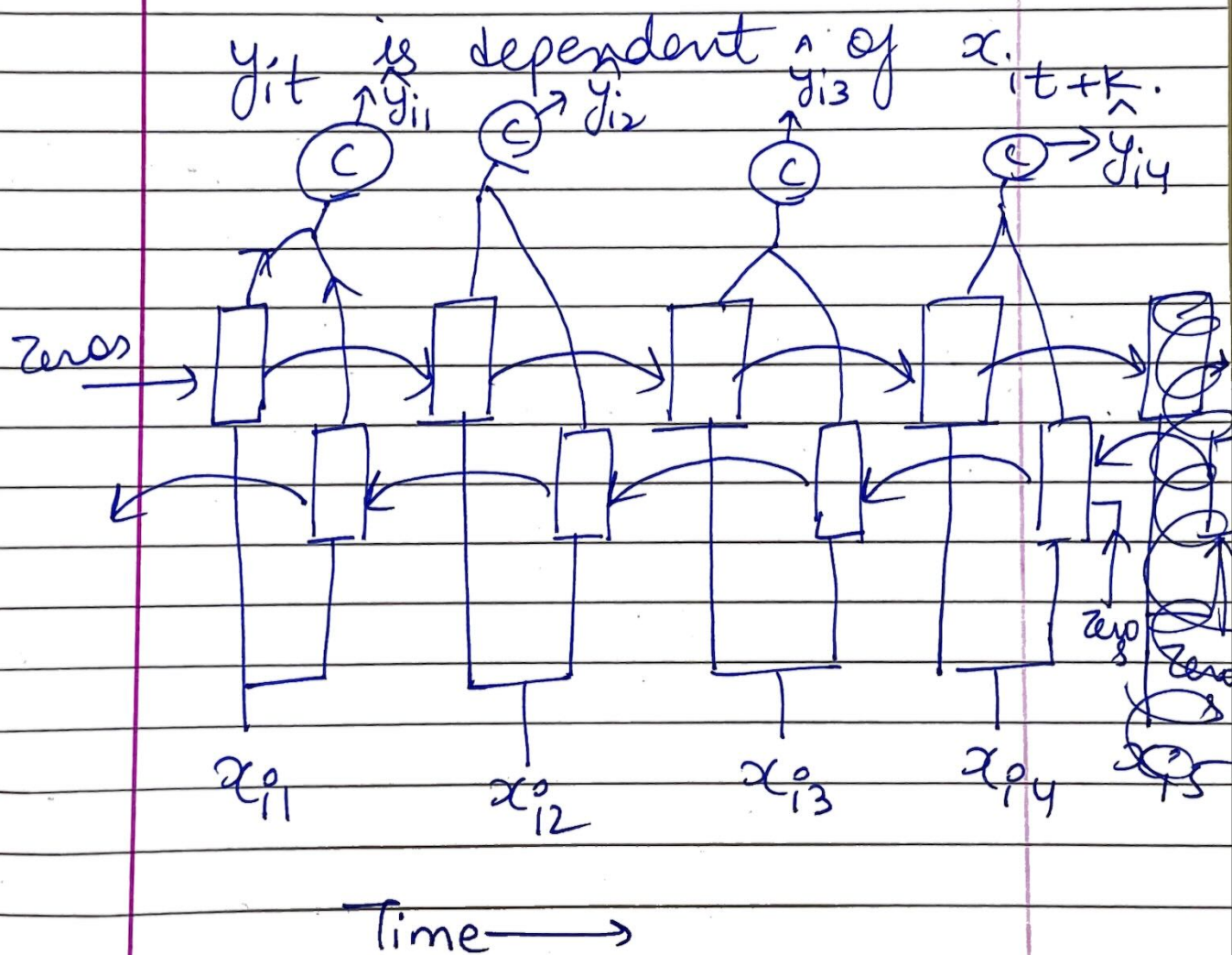
Date :

Page No. :

V9: Bidirectional RNN



Q What if y_{i3} is dependent on x_{i5}



Two directions of connections. of two types

Speech - Audio (first recording should complete to have bi-RNN)

Speech - Audio (first recording should complete to have bi-RNN)